

Le Yu

Senior Machine Learning Engineer | Ads Modeling & Bidding

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Experience

TikTok, Senior Machine Learning Engineer

Jan 2025 – present

App Ads Bidding

Beijing, China

San Jose, CA

- **Delayed GMV prediction:** Drove the end-to-end design and launch of a **hybrid delayed-GMV model** for Target ROAS bidding, fusing delay curves with click and real-time funnel signals to predict final GMV before feedback matured. The initial IAA launch improved advertiser value by about 7%.
- **Long-horizon bid control:** Designed and launched a **multi-day MPC** controller that incorporated historical spend and predicted GMV into each bidding decision to correct accumulated delivery gaps. An IAP experiment increased the share of campaigns meeting ROAS targets by about 14% and advertiser value by about 37%.
- **Generative bidding:** Built minute-level state-action-reward trajectories, RTG targets, and training and serving pipelines for sequence-based bidding. Prototyped **Decision Transformer** and generator-critic-value models with controller-based action bounds and safety priors.
- **Ads diagnosis agent:** Designed and launched a production App Ads diagnosis agent for ticket parsing, funnel triage, and bidding and ranking investigations. Authored its modular Skills and integrated internal tools with evidence and confidence controls to deliver traceable diagnoses while separating verified findings from hypotheses.

TikTok, Machine Learning Engineer

Mar 2023 – Dec 2024

Lead Ads Ranking

Beijing, China

- **Federated learning framework: PACS-FL** keeps privacy-sensitive conversion labels in a protected sandbox, exchanging intermediate tensors and gradients with cloud workers for split training. Drove Framework 2.0 for Lead Ads: a C++/Proto training interface with a unified label-feature pipeline, validated automatic cloud-sandbox graph partitioning, and single-job orchestration across GPU batch, CPU batch, and streaming stages, cutting PS storage by 70% and batch time by 90%.
- **Unified conversion model:** Drove the staged unification of trackable and privacy-restricted traffic across on- and off-platform Lead Ads into one multi-head model with shared training and serving. Traffic-conditioned adapters and a compact sparse-feature tower preserved segment-specific behavior. The broader **One Model** program shipped through 18 production launches across US, EU, and ROW, with individual unification launches delivering up to 18.6% advertiser-value lift
- **Privacy-compliant ad retrieval:** Launched multiple **recall branches** using complementary privacy-safe signals: ELAND landing-page interaction graphs for U2I matching; Memory Recall for cluster-based candidate transfer; Cohort and DP behavioral features; and PACS-FL pCVR for eCPM retrieval.

TikTok, Machine Learning Engineer

July 2021 – Feb 2023

Promotion Ads Ranking

Beijing, China

- **Ads-recommendation collaboration:** Built bidirectional sample and feature pipelines between Promote and TikTok's recommendation system, allowing promotion models to use organic engagement signals and recommendation models to learn from paid-promotion feedback.
- **New ad product launches:** Launched Promote Internal Ads, Ad4Ad, and Promote-to-Live, building their CVR models and conversion pipelines. For Promote-to-Live, replaced daily updates with streaming training and modeled delayed watch-time labels.

Education

Peking University, Master

Sept 2018 – July 2021

Electronics and Communication Engineering

Beijing

National TsingHua University, Exchange

Feb 2016 – June 2016

Electronic Engineering

Hsinchu

Southeast University, Bachelor

Sept 2014 – July 2018

Automation Engineering

Nanjing

Internships

E Fund Management, Machine Learning Intern

Nov 2020 – Nov 2020

- Built **LSTM** baselines over hundreds of proprietary market features for equity time-series forecasting.

Microsoft, Machine Learning Intern

July 2020 – Sept 2020

- Fine-tuned **BERT** to classify articles for Bing News and support personalized newsletter curation.

WeChat, Machine Learning Intern

Apr 2020 – June 2020

- Developed a **ResNet**-based visual model for detecting unauthorized and reposted videos on WeChat Channels.

Tencent, Strategy Intern

Aug 2019 – Nov 2019

- Analyzed global cloud-computing markets and competitors to support Tencent Cloud product planning.

Microsoft, Machine Learning Intern

Mar 2019 – July 2019

- Improved an **N-gram language model** for first-page result reranking in Bing Search.

Skills

Machine Learning: Ads ranking and intelligent bidding, conversion-rate and value prediction, delayed GMV prediction, privacy-preserving ML, sequential decision modeling

Engineering: C++, Python, Go, SQL, TensorFlow, Spark, distributed training, real-time serving, streaming data pipelines

Languages: Mandarin Chinese, English